Sex-differences in Morbidity Burden

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Abstract

Women live longer than men yet carry a greater burden of multimorbidity across the adult life course, a paradox that remains poorly characterized in contemporary diverse populations. Using the NIH *All of Us* Research Program (n = 344,038), we examined sex differences in mean morbidity count across ten age intervals from 18 to 125 years. Women had significantly more morbidities than men across all age groups (differences: 5.58–8.93; $p < 0.01$). The disparity was largest during midlife (40–49 years, difference = 8.93), coinciding with the perimenopausal and menopausal transitions, and persisted into the oldest age groups. These findings demonstrate that women experience a shorter health span than men despite greater longevity, underscoring the need for greater recognition, research funding, and clinical trial representation focused on women's health.

Background

Men have shorter life expectancies than women. In addition, a sex specific "morbidity–mortality paradox" has been known for over five decades, although research has neither explained its causes nor narrowed this gap. Quantification of the multimorbidity burden may inform efforts to optimize health span, especially for women.

Methods

The sample for this study included individuals age 18–125 years without a recorded death in the NIH *All of Us* Research Program. Analyses were limited to morbidities with ≥1,000 occurrences that were clinically recorded in the electronic medical record with a systematized nomenclature of medicine code (SNOMED). Age was categorized into intervals: 18–29, 30–39, 40–49, 50–59, 60–69, 70–79, 80–89, 90–99, 100–109, 110–125 years. The primary outcome was the difference in mean number of morbidities between women and men across age intervals and were assessed using Welch's t-test to account for unequal variances and group sizes using Python 3.10.12.

Results

The analytic sample included 344,038 adults (n=215,645 women, n=128,393 men). Table 1 displays the mean number of morbidities by sex across age intervals. Figure 1 depicts the difference in age-related trends. Multimorbidity increased progressively with age for both sexes. However, across all age categories, women had a significantly higher morbidity burden than men, with differences ranging from 5.58 to 8.93 additional morbidities (p<0.01). In early adulthood (18–29 years), women had a mean of 5.96 more morbidities than men (20.88; 14.92; t = 32.78; p<0.01). The disparity in morbidity burden widened substantially during midlife. The largest disparity was detected among individuals aged 40–49 years, when women had 8.93 more morbidities than men (35.05; 26.12; t = 52.21; p<0.01), and 50–59 with women experiencing 8.92 additional morbidities relative to men (42.10; 33.19; t = 51.34; p<0.01). Although the absolute magnitude of differences narrowed slightly in older adulthood, females continued to have significantly higher mean values (60–69 years, difference, 7.70; 70–79 years, difference 6.12 and 80–89 years, difference 5.64). The overall morbidity burden peaked for both sexes at ages 90–99 years, a decade when women had 5.58 more morbidities than men (53.86; 48.28, t = 28.61; p<0.01). In age groups >100, the same patterns of significantly higher morbidities in women persist.

Discussion

These data from a contemporary diverse cohort demonstrate a consistent pattern in which women have larger, and steeper accumulation of multimorbidity across the adult life course relative to men. Additionally, the concentration with the greatest disparity in multimorbidity during midlife coincides with the understudied perimenopausal and menopausal transitions. The cause for differential multimorbidity may be multifactorial, including differences in healthcare seeking behaviors or differing health care during reproductive events, such as pregnancy. Other factors may increase vulnerability to the development and accumulation of additional morbidities, including differential exposure to social and environmental stressors such as higher domestic violence[1], caregiving responsibilities[2] and psychosocial stress.[3] Rigorous efforts to address the lower health span in women are needed to addressed historical lack of recognition, measurement, representation in clinical trials[4] and funding for women's health.

Conclusion

Between ages 18–80 years, women have a consistently higher morbidity burden than men. Women experience a shorter health span than men with stark differences in multi-morbidity, especially in midlife.

Table 1: Multimorbidity by Age and Sex

Age Interval	Mean # Morbidities (Male)	Mean # Morbidities (Female)	Difference (Female−Male)	t-Statistic	p-Value
18–29	14.92	20.88	5.96	32.78	<.001
30–39	20.76	28.72	7.96	44.98	<.001
40–49	26.12	35.05	8.93	52.21	<.001
50–59	33.19	42.10	8.92	51.34	<.001
60–69	41.06	48.76	7.70	42.53	<.001
70–79	46.70	52.81	6.12	32.03	<.001
80–89	48.14	53.78	5.64	28.96	<.001
90–99	48.28	53.86	5.58	28.61	<.001
100–109	48.28	53.86	5.58	28.61	<.001
110–125	48.28	53.86	5.58	28.61	<.001

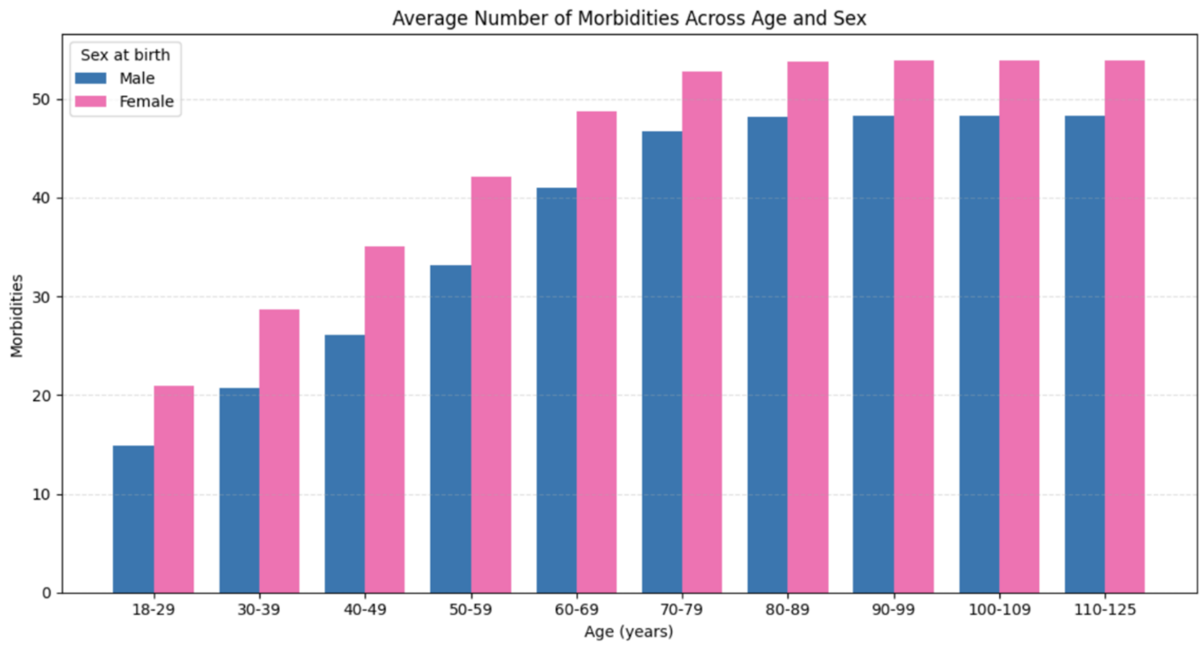

Figure 1: Average Morbidities by Age and Sex

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